

Bili-Study Cache Lab: Orchid Protocol

An authored learning handout for validating document retrieval. Lesson date: 2026-10-07. Orchid Protocol is the name of this classroom exercise, not an industry standard.

The Orchid Protocol exercise

For this exercise, each cache key has a version number. A database update increases the version number. A reader only accepts a cached value whose version matches the current database version. The class chooses a TTL of 37 seconds as an arbitrary exercise parameter. TTL bounds age; it does not prove that a value is consistent with a concurrent database update.

Single-flight on a cache miss

When many readers miss the same key, single-flight lets one request fetch the value while the other requests wait for that result. This reduces duplicate database work for that key. A per-key lock must have a bounded timeout so a crashed loader cannot block readers forever.

Failure and practice

If the cache is unavailable, read the database directly with appropriate rate limits. Never treat cache failure as proof that data is absent. Practice: explain why the 37-second TTL alone cannot implement the version check, and why single-flight controls duplicate work rather than consistency.